PSWC – Practice Questions

Question 1:

A university checks scholarship eligibility based on family income and entrance exam score. Write a C program using user-defined functions to determine whether a student is eligible.

Eligibility Criteria:
Family income must be at most Rs. 5,00,000
Entrance score must be at least 85

Question 2:

A barcode validation system verifies whether a product's numeric barcode reads the same forwards and backwards. Write a C program that reverses a given positive integer using a while loop and reports whether the original number equals its reverse. Conversion to string is not permitted.

Question 3:

A mobile service provider calculates the monthly internet data bill based on data usage. The first 10 GB costs Rs. 100. For every additional GB, Rs. 20 is charged. Write a C program using user-defined functions to calculate and display the total data bill.

Question 4:

Perform selection sort in descending order for a given set of inputs.

Question 5:

Using selection sort, sort only even elements in ascending order; keep odd numbers in the same position.

Question 6:

Define a structure HotelBooking with members: bookingId, guestName, roomCost, and hoursBeforeCheckIn.

Refund rules:
>= 48 hours: 90% refund
24 to 47 hours: 75% refund
12 to 23 hours: 50% refund
< 12 hours: No refund

Display refund amount for each booking and total refund amount.

Question 7:

Define a structure Vehicle with members: vehicleNumber, vehicleType, and hoursParked.

Parking fee rules:
  Type S: First 3 hrs Rs. 30, additional Rs. 15/hr
  Type M: First 3 hrs Rs. 15, additional Rs. 8/hr
  Type L: First 3 hrs Rs. 50, additional Rs. 25/hr

Display parking fee for each vehicle and the vehicle that paid the highest fee.

Question 8:

A project management dashboard converts a project's total duration given in days into an equivalent breakdown of years, months, and remaining days for display on the timeline view. Assuming 1 year = 365 days and 1 month = 30 days, write a C program that reads the total number of days and prints the equivalent in years, months, and days using only integer division (/) and modulus (%) operators.

Question 9:

A sensor logger initially records readings for a few time slots but later needs to append more entries. The system should dynamically expand storage without losing existing data.

Write a C program using realloc():
- Allocate memory for 3 integer readings.
- Read the first 3 readings.
- Expand the array to hold 5 readings.
- Read 2 additional readings.
- Display all 5 readings.
- Calculate and display the average.

Question 10:

An IoT firmware engineer wants to understand how different memory allocation methods behave before sensor data is written.

Write a C program to compare memory allocated using malloc() and calloc():
- Allocate an array of 5 integers using malloc().
- Allocate another array of 5 integers using calloc().
- Print all elements of both arrays without initializing them.
- Observe the difference in values.
- Assign values to the calloc array.
- Compute and display the product of all elements in the calloc array.

Question 11:

Define a structure Equipment with members: equipId, name, daysOverdue, and category.

Fine rules:
  Category M: Rs. 5/day

Category E: Rs. 10/day
Category T: Rs. 2/day

Write a program to read details of n equipment items and calculate the fine for each. Also display the item with the highest fine.

Question 12:

Write a C program to reverse a 1D array in-place using pointer arithmetic.

Question 13:

Count the number of swaps performed in selection sort for a given array of integers.

Question 14:

A network security module needs to verify whether a candidate number is prime before using it as a session token seed. Write a C program that determines whether a given positive integer N is prime or not.

Question 15:

Given an employee ID list, search for an ID using recursive linear search. Return index if found, else -1.

Question 16:

Define a structure EmergencyCall with members: name, age, distanceKm, and urgencyScore.

Dispatch priority is calculated as:
Priority = urgencyScore + (50 - distanceKm)
If age is above 65, add 10 more priority points.

Sort calls in decreasing order of priority.

Question 17:

A warehouse has multiple rows, and each row contains 3 storage bins identified by bin numbers. Write a C program to display all bin numbers using a pointer to an array.

Question 18:

A scientific laboratory console repeatedly accepts a researcher's choice from a menu and performs the requested arithmetic operation on two operands. Operations supported: 1) Add, 2) Subtract, 3) Multiply, 4) Divide, 5) Exit. Implement the dispatch logic using switch-case inside a do-while loop. Division by zero must be handled gracefully without terminating the program.

Question 19:

Define a structure Satellite with members: satelliteId, batteryLevel, orbitAltitude, and payloadMass.

A satellite is flagged if:
- batteryLevel < 25, OR
- payloadMass > 500, OR
- orbitAltitude > 2000

Write a function that accepts a pointer to a structure and returns the safety flag status.

Question 20:

A file manager receives 6 file sizes (in KB). Write a C program to find the maximum file size, minimum file size, and the difference between them using a pointer to array.

Question 21:

A recreational mathematics platform highlights Armstrong (narcissistic) numbers within a user-defined range for educational puzzles. A 3-digit Armstrong number equals the sum of its digits each raised to the power 3; for example, $153 = 1^3 + 5^3 + 3^3$. Write a C program that prints all 3-digit Armstrong numbers within a given range [L, R] using nested loops.

Question 22:

Using binary search, find the first position where element ≥ key in a sorted exam score list and return its value.

Question 23:

An athlete has recorded scores in 4 track events. Write a C program to find the highest score using a pointer to array.

Question 24:

A sports academy calculates the total score, average score, and grade of an athlete based on performance in 5 events. Write a C program using user-defined functions to calculate and display the result.

Grade Rules:
- Average ≥ 90: A
- Average ≥ 75: B
- Average ≥ 60: C
- Average ≥ 40: D
- Below 40: Fail

Question 25:

An astronomy software needs to correctly mark leap years to align planetary observation schedules on February 29. A year is a leap year if it is divisible by 4 but not by 100, unless it is also divisible by 400. Write a C program using nested if-else constructs to determine whether a given year is a leap year.

Question 26:

Define a structure FarmProduct with members: productName, cultivationCost, marketPrice, and quantityKg.

Profit = marketPrice * quantityKg - cultivationCost

Display profit or loss for each product. Also display the product with maximum profit.

Question 27:

A compiler tool kit is developing a lightweight string handling module to replace built-in string functions for an embedded target.

Write a C program with user-defined functions for string operations:
- Define a function to copy one string to another.
- Define a function to calculate the length of a string.
- Define a function to compare two strings.
- In main(), read two strings from the user.
- Call all three functions and display the results.

Question 28:

Define a structure CityTemperature with members: cityName, temperature[7].

For each city, calculate total and average temperature for 7 days. Classify heat risk as:
Average temperature >= 40°C: High Risk
Average temperature >= 30°C: Moderate Risk
Otherwise: Low Risk

Display each city's risk category and the city with maximum total temperature.

Question 29:

Two quiz platforms generate score streams of equal size. Write a C program to merge the two streams into a third array by adding the corresponding elements using pointers to arrays.

Question 30:

A startup HR system generates the salary slip of a software developer. Basic salary is entered by the user. Write a C program using user-defined functions to calculate Performance Allowance (PA), Work-from-Home Stipend (WHS), Pension Fund (PensF), and net salary.

Rules:
PA = 40% of basic
WHS = 20% of basic
PensF = 12% of basic
Net Salary = Basic + PA + WHS − PensF

Question 31:

A game engine renders a triangular score board to illustrate nested loop rendering. Write a C program that prints Floyd's Triangle of N rows, in which consecutive natural numbers are arranged so that row i contains exactly i numbers. Use nested loops.

Question 32:

An e-learning platform offers discounts on course bundles based on the total purchase amount. Write a C program using user-defined functions to calculate the discount and final payable amount.

Discount Rules:

- Amount ≥ Rs. 5000: 20% discount
- Amount ≥ Rs. 3000: 10% discount
- Amount below Rs. 3000: 5% discount

Question 33:

A digital signal processing library uses Fibonacci numbers to generate test waveforms. Write a C program that prints the first N terms of the Fibonacci sequence, where each term is the sum of the two preceding terms (starting with 0 and 1). Use a for loop. Recursion is not permitted.

Question 34:

Define a structure Clinic with members: name, appointmentsHandled, totalRating, and lateArrivals.

Calculate the average rating of each clinic. A clinic is marked Recommended if: average rating >= 4.0 and lateArrivals <= 2.

Display all recommended clinics and the clinic with the highest average rating.

Question 35:

A drone serial number verification system checks authenticity by reversing the digits of a 3-digit numeric code. Given a 3-digit positive integer N, write a C program that reverses its digits using only the modulus (%), integer division (/), addition, and multiplication operators. Loops, recursion, arrays, and library functions are not permitted.

Question 36:

A food delivery application calculates the delivery fee based on distance from the restaurant. Write a C program using user-defined functions to read the distance, calculate the fee, and display the total.

Fee Rules:

- First 3 km: Rs. 30 fixed
- Next 7 km: Rs. 8/km
- Above 10 km: Rs. 12/km

Question 37:

A survey platform needs to compute statistics for response scores collected at runtime. The number of respondents is not fixed in advance.

Write a C program using dynamic memory allocation:
- Read the number of elements n.
- Dynamically allocate memory for n integers.
- Read n integer values from the user.
- Calculate the sum of all elements.
- Calculate the average.
- Display the sum and average.

Question 38:

A hospital manages patient records dynamically, allowing admission of new patients at runtime. The system must identify the most critical patient (highest severity score) and display sorted records.

Write a C program using structures and dynamic memory allocation:
- Define a structure with name, patient ID, and severity score.
- Allocate memory for n patients.
- Read patient details.
- Allow the user to add more patients dynamically.
- Find the patient with highest severity score.
- Sort all patients based on severity score.
- Display the most critical patient and sorted list.

Question 39:

A task queue manager stores job IDs dynamically using a linked list. The system must support insertion, deletion, and display operations.

Write a C program to implement a singly linked list:
- Define a node structure containing an integer and a pointer.
- Insert elements at the end of the list.
- Display all elements in the list.

Question 40:

An image processing utility simplifies the aspect ratio of a display by computing the GCD of the width and height in pixels. Write a C program that computes the GCD of two positive integers using the iterative Euclidean algorithm with a while loop. Recursion and library functions are not permitted.

Question 41:

A social media post analyzer studies the composition of hashtags entered by users. The system determines how many vowels and consonants are present in a given string without relying on built-in string functions.

Write a C program to perform this analysis using pointer traversal:

- Read a string from the user.
- Traverse the string using a character pointer.
- Count the number of vowels (a, e, i, o, u) in the string.
- Count the number of consonants in the string.

Question 42:

Perform selection sort in ascending order for a given set of inputs.

Question 43:

Given a sorted list of altitude readings, find the index of a given altitude using recursive binary search.

Question 44:

A username validation system checks whether a chosen username reads the same forwards and backwards as part of a novelty filter. The comparison must be done manually without built-in functions.

Write a C program to check whether a string is a palindrome:

- Read a string from the user.
- Use two pointers — one from the beginning and one from the end.

Question 45:

A conveyor belt controller stores 6 slot identifiers. Write a C program to perform a circular right shift by one position using a pointer to array, where the last slot identifier moves to the first position.

Question 46:

A digital library portal allows a member to check out books only if the number of requested books does not exceed the available copies and is within the member's borrowing limit (a multiple of 1). Write a C program using user-defined functions to validate the checkout and display the updated available copies.

Question 47:

Define a structure Intern with members: name, weeklyTaskScore, technicalTestScore, presentationScore.

Final Score = 0.4 * weeklyTaskScore + 0.2 * technicalTestScore + 0.4 * presentationScore

An intern earns a Pre-Placement Offer (PPO) if final score is at least 50. Display PPO eligibility and topper details.

Question 48:

A weather station stores the last 5 humidity readings (in %) in a one-dimensional array. Write a C program to access the entire array using a pointer to array, display all the readings, and compute the average humidity.

Question 49:

An EV charging station calculates the time required to charge a battery from current level to full capacity. Write a C program using user-defined functions to calculate the charging time.

Formula: Time = (Battery Capacity − Current Level) / Charging Rate

Question 50:

Define a structure MedicineStock with members: medicineId, medicineName, currentStock, weeklyDemand.

A medicine must be reordered if: currentStock < 0.25 * weeklyDemand
Suggested reorder quantity: weeklyDemand - currentStock

Display medicines requiring reorder.

Question 51:

A stadium parking system calculates the parking fee based on the number of hours a vehicle is parked. Write a C program using user-defined functions to calculate and display the parking fee.

Fee Rules:
First 3 hours: Rs. 50 fixed
Every additional hour: Rs. 25/hour

Question 52:

A marathon timing system stores finish times in seconds in an unsorted array. Write a C program to sort the times in ascending order using selection sort and a pointer to array.

Question 53:

A courier company's tracking system derives a single verification digit from each shipment's tracking number by summing all its digits. Write a C program that reads a positive integer and computes the sum of its digits using a while loop. String operations are not allowed.

Question 54:

An app store records the daily downloads of 3 apps for a given number of days. Write a C program to calculate and display the total downloads for each day using a pointer to array.

Question 55:

A cloud resource allocator must safely handle dynamic memory and avoid crashes due to memory errors. It should also dynamically grow storage as needed.

Write a C program implementing safe memory handling:

- Create a function that safely allocates memory and handles errors (Use errno and perror/strerror).
- Create a function that safely reallocates memory.
- Start with an array of capacity 4.
- Read integers from the user until -1 is entered.
- Expand the array dynamically when capacity is reached.
- Display all entered elements and total count.

Question 56:

A DNA sequence processing tool needs to reverse a nucleotide string for complementary strand analysis. The reversal must be done efficiently without using extra memory.

Write a C program to reverse a string using the two-pointer technique:

- Read a string from the user.
- Use one pointer at the beginning and another at the end of the string.
- Swap characters until the pointers meet.
- Perform the reversal in-place.
- Display the reversed string.

Question 57:

An online auction platform determines whether a seller made a profit or a loss on an item based on the reserve price (RP) and final bid price (BP), along with the corresponding percentage. Write a C program that reads RP and BP and prints the transaction type and percentage. Use exactly one ternary expression to choose between "Profit" and "Loss". Assume RP ≠ BP. Use of if-else is not permitted.

Question 58:

Find the minimum delivery time (in minutes) in a list using linear search.

Question 59:

Find the first and last position of an order ID in the dispatch log using linear search.

Question 60:

A fitness app calculates the Hydration Index (HI) of a person using daily water intake (litres) and body weight (kg). Write a C program using user-defined functions to calculate HI and display the hydration category.

Formula: HI = water_intake / (weight * 0.033)

Category Rules:

HI < 0.8: Dehydrated

0.8 to 1.0: Adequate

1.0 to 1.2: Well Hydrated
HI > 1.2: Over Hydrated

Question 61:

An audio mixer represents a 2D matrix where each element holds a channel's volume level. Write a C program to increase every volume level by 5 using a pointer to array.

Question 62:

An IT company's payroll portal computes the net monthly take-home pay of an engineer from the basic pay using these rules: City Allowance (CA) is 40% of the basic pay, Transport Allowance (TA) is 20% of the basic pay, and Employee Provident Fund (EPF) deduction is 12% of the basic pay. The net pay is given by Net = Basic + CA + TA − EPF. Write a C program that reads the basic pay and prints the net pay using only arithmetic operators.

Question 63:

A logistics system stores sorted delivery times in minutes. Write a C program to search for a given delivery time using binary search and a pointer to array. Display the index if found; otherwise, display "Not found".

Question 64:

Using Linear search, find how many times a given product ID appears in the inventory log.

Question 65:

A cybersecurity audit tool lists all prime numbers in a given port range to flag potentially insecure open ports used by legacy protocols. Write a C program that prints every prime number using nested loops.